

CS228 : Logic for Computer Science

Module 1: Propositional Logic

Lecture 5: Soundness and Completeness of Natural Deduction

Instructor: S. Akshay

IIT Bombay, India

Recap: Rules of Natural Deduction

Conjunction:

$$\frac{F \quad G}{F \wedge G} \wedge_i \quad \frac{F \wedge G}{F} \wedge_{e1} \quad \frac{F \wedge G}{G} \wedge_{e2}$$

Double Negation:

$$\frac{\neg\neg F}{F} \neg\neg_e \quad \frac{G}{\neg\neg G} \neg\neg_i$$

Implication (MP, MT & Intro):

$$\frac{F \quad F \rightarrow G}{G} \text{MP} \quad \frac{F \rightarrow G \quad \neg G}{\neg F} \text{MT} \quad \frac{\boxed{\begin{array}{c} F \\ \vdots \\ G \end{array}}}{F \rightarrow G} \rightarrow_i$$

Disjunction:

$$\frac{F}{F \vee G} \vee_{i1} \quad \frac{G}{F \vee G} \vee_{i2} \quad \frac{F \vee G \quad \boxed{\begin{array}{c} F \\ \vdots \\ H \end{array}} \quad \boxed{\begin{array}{c} G \\ \vdots \\ H \end{array}}}{H} \vee_e$$

Single Negation Rules

- ▶ We use **contradiction**, an expression of the form $F \wedge \neg F$, for a propositional formula F .
- ▶ Two contradictions are equivalent: $p \wedge \neg p$ is equivalent to $\neg r \wedge r$, so we will use $\perp$!

$$\frac{\perp}{F} \perp e$$

$$\frac{F \quad \neg F}{\perp} \neg e$$

$$\frac{\boxed{\begin{array}{c} F \\ \vdots \\ \perp \end{array}}}{\neg F} \neg i$$

Exercise 5.1: Derive $p \rightarrow \neg p \vdash \neg p$.

1.	$p \rightarrow \neg p$	premise
2.	p	assumption
3.	$\neg p$	MP 1,2
4.	$\perp$	$\neg e$ 2,3
5.	$\neg p$	$\neg i$ 2-4

Some Derived Rules

Exercise 5.2

- (i) Prove that Modus Tollens rule can be derived from the other rules above! Are there any other such rules that can be derived? Hint: what about $\neg\neg i$?
- (ii) Prove “proof by contradiction”, i.e., show that if from $\neg F$ we can derive $\perp$, then we can deduce F . Hint: use only $\rightarrow$ intro/elim rules and negation intro/double negation elimination.
- (iii) Prove the theorem called Law of Excluded Middle (also called LEM): $\vdash F \vee \neg F$ is valid!

Exercise 5.2(i)-(ii): MT and Proof by Contradiction

(i) $F \rightarrow G, \neg G \vdash \neg F$

- | | | |
|----|-------------------|--------------|
| 1. | $F \rightarrow G$ | premise |
| 2. | $\neg G$ | premise |
| 3. | F | assumption |
| 4. | G | MP 1,3 |
| 5. | $\perp$ | $\neg e$ 4,2 |
| 6. | $\neg F$ | $\neg i$ 3-5 |

(ii) $\neg F \rightarrow \perp \vdash F$

- | | | |
|----|----------------------------|------------------------------|
| 1. | $\neg F \rightarrow \perp$ | given (via $\rightarrow i$) |
| 2. | $\neg F$ | assumption |
| 3. | $\perp$ | MP 1,2 |
| 4. | $\neg\neg F$ | $\neg i$ 2-3 |
| 5. | F | $\neg\neg e$ 4 |

Exercise 5.2(iii): Law of Excluded Middle (LEM)

$$\vdash F \vee \neg F$$

1.	$\neg(F \vee \neg F)$	assumption
2.	F	assumption
3.	$F \vee \neg F$	$\vee i_1$ 2
4.	$\perp$	$\neg e$ 3,1
5.	$\neg F$	$\neg i$ 2-4
6.	$F \vee \neg F$	$\vee i_2$ 5
7.	$\perp$	$\neg e$ 6,1
8.	$\neg\neg(F \vee \neg F)$	$\neg i$ 1-7
9.	$F \vee \neg F$	$\neg\neg e$ 8

Summary: All Rules of Natural Deduction

Conjunction:

$$\frac{F \quad G}{F \wedge G} \wedge i$$

$$\frac{F \wedge G}{F} \wedge e_1$$

$$\frac{F \wedge G}{G} \wedge e_2$$

Disjunction:

$$\frac{F}{F \vee G} \vee i_1$$

$$\frac{G}{F \vee G} \vee i_2$$

$$\frac{F \vee G \quad \boxed{\begin{array}{c} F \\ \vdots \\ H \end{array}} \quad \boxed{\begin{array}{c} G \\ \vdots \\ H \end{array}}}{H} \vee e$$

Double Negation:

$$\frac{\neg\neg F}{F} \neg\neg e$$

Implication:

$$\frac{F \quad F \rightarrow G}{G} \text{MP}$$

$$\frac{\boxed{\begin{array}{c} F \\ \vdots \\ G \end{array}}}{F \rightarrow G} \rightarrow i$$

Negation & Bot:

$$\frac{\boxed{\begin{array}{c} F \\ \vdots \\ \perp \end{array}}}{\neg F} \neg i$$

$$\frac{F \quad \neg F}{\perp} \neg e \quad \frac{\perp}{F} \perp e$$

Derived Rules:

$$\frac{F \rightarrow G \quad \neg G}{\neg F} \text{MT}$$

$$\frac{\boxed{\begin{array}{c} \neg F \\ \vdots \\ \perp \end{array}}}{F} \text{PBC}$$

$$\frac{G}{\neg\neg G} \neg\neg i$$

$$\text{LEM: } \vdash F \vee \neg F$$

Topic 5.1

Soundness of Natural Deduction for Propositional Logic

Soundness of Natural Deduction for Propositional Logic

Theorem (What is derivable is true!)

If $F_1, \dots, F_n \vdash G$ is valid, then $F_1, \dots, F_n \models G$ holds

i.e., G being provable from $F_1, \dots, F_n$ means: for every valuation, whenever $F_1, \dots, F_n$ evaluate to 1, G also evaluates to 1.

Proof: By Induction (on what?!)

- ▶ Assume $F_1, \dots, F_n \vdash G$.
- ▶ Thus there is some proof that yields G . Induction is on number of lines, k , of this proof!
- ▶ When $k = 1$, there is only one line in the proof, which is the premise, say F . This gives the sequent $F \vdash F$. But in this case $F \models F$!
- ▶ Induction Hypothesis: Whenever $F_1, \dots, F_n \vdash G$ is derived using a proof of length $\leq k - 1$, we have $F_1, \dots, F_n \models G$.

Note! We are using “Strong Induction”

Soundness of Natural Deduction for Propositional Logic (Contd.)

- ▶ Consider a proof with k lines.
- ▶ How did we arrive at G ? Which proof rule gave G as the last line? **Case Analysis!**
 - ▶ Case 1: Assume G was obtained using $\wedge i$ rule.
 - ▶ Then G is of the form $G_1 \wedge G_2$.
 - ▶ G_1 and G_2 were proved earlier, say at lines $k_1, k_2 < k$.
 - ▶ i.e., we have shorter proofs $F_1, \dots, F_n \vdash G_1$ and $F_1, \dots, F_n \vdash G_2$
 - ▶ By inductive hypothesis, $F_1, \dots, F_n \models G_1$ and $F_1, \dots, F_n \models G_2$. This implies, $F_1, \dots, F_n \models G_1 \wedge G_2$ (by definition!).
 - ▶ Case 2: Assume G was obtained using $\rightarrow i$, i.e., G is of the form $G_1 \rightarrow G_2$.
 - ▶ A box starting with G_1 was opened at some line $k_1 < k$.
 - ▶ Last line in box is G_2 and line just after the box is G .
 - ▶ Add G_1 as premise, along with $F_1, \dots, F_n$, i.e., get proof $F_1, \dots, F_n, G_1 \vdash G_2$, of length $k - 1$. By inductive hypothesis, $F_1, \dots, F_n, G_1 \models G_2$. But, this is same as $F_1, \dots, F_n \models G_1 \rightarrow G_2$
 - ▶ All other cases are by considering each rule separately.
 - ▶ **Exercise 5.4: Check all other 10 cases and complete the proof!**

□

Topic 5.2

Completeness of Natural Deduction for Propositional Logic

Completeness of Natural Deduction for propositional logic

Theorem (Anything that is correct is provable)

If $F_1, \dots, F_n \models G$ holds, then $F_1, \dots, F_n \vdash G$ is valid

Whenever $F_1, \dots, F_n$ semantically entail G , then G can be derived from $F_1, \dots, F_n$ using the ND rules.

Proof: Let $F_1, \dots, F_n \models G$. Proof is in three steps

- ▶ Step 1: Show that $\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ holds. This is easy!
 - ▶ By contradiction, suppose $\not\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$.
 - ▶ This is a nested implication, is false only if there exists an assignment s.t. G evaluates to 0 and $F_1, \dots, F_n$ evaluate to 1.
 - ▶ Which contradicts $F_1, \dots, F_n \models G$.
- ▶ Step 2: Show that $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ is valid
- ▶ Step 3: Show that $F_1, \dots, F_n \vdash G$ is valid

Completeness of Natural Deduction for propositional logic

Theorem (Anything that is correct is provable)

If $F_1, \dots, F_n \models G$ holds, then $F_1, \dots, F_n \vdash G$ is valid

Whenever $F_1, \dots, F_n$ semantically entail G , then G can be derived from $F_1, \dots, F_n$ using the ND rules.

Proof: Let $F_1, \dots, F_n \models G$. Proof is in three steps

- ▶ Step 1: Show that $\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ holds.
- ▶ Step 2: Show that $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ is valid
- ▶ Step 3: Show that $F_1, \dots, F_n \vdash G$ is valid **This is also easy!**
 - ▶ Take the proof from Step 2.
 - ▶ Add $F_1, \dots, F_n$ as premises as done in soundness proof.
 - ▶ Then apply *MP* n times to arrive at G .

Completeness of Natural Deduction for propositional logic

Theorem (Anything that is correct is provable)

If $F_1, \dots, F_n \models G$ holds, then $F_1, \dots, F_n \vdash G$ is valid

Whenever $F_1, \dots, F_n$ semantically entail G , then G can be derived from $F_1, \dots, F_n$ using the ND rules.

Proof: Let $F_1, \dots, F_n \models G$. Proof is in three steps

- ▶ Step 1: Show that $\models F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ holds.
- ▶ Step 2: Show that $\vdash F_1 \rightarrow (F_2 \rightarrow (\dots (F_n \rightarrow G) \dots))$ is valid
- ▶ Step 3: Show that $F_1, \dots, F_n \vdash G$ is valid

So, the only thing left is step 2 from step 1. This is the hard part!

Completeness of Natural Deduction for propositional logic

Step 2 follows from the following theorem:

Theorem

For any propositional formula G , if $\models G$ holds, then $\vdash G$ is valid. That is, if G is a tautology, then it is a theorem!

Proof

- ▶ Let $p_1, \dots, p_m$ be the propositional variables in G ; all 2^m valuations make G true.
- ▶ From just this, can derive a proof of G ?
 - ▶ Step 2.1 **Key Idea**: Encode each line of truth table as a sequent!
 - ▶ If in line ℓ of truth table, p_i has value 1, then set $\hat{p}_i = p_i$ otherwise set $\hat{p}_i = \neg p_i$
 - ▶ Sequent for line ℓ : $\hat{p}_1 \dots \hat{p}_m \vdash G$
 - ▶ Step 2.2: Construct proofs for each of 2^m sequents.
 - ▶ For each line ℓ show that the sequent $\hat{p}_1 \dots \hat{p}_m \vdash G$ is valid.
 - ▶ Step 2.3: Assemble them to obtain proof of G

Completeness Proof contd

Lemma

Let F be any propositional formula with variables $p_1, \dots, p_m$. Let ℓ be a line number in Truth Table of F . Let $\hat{p}_i = p_i$ if p_i has value 1 in line ℓ , and $= \neg p_i$ if p_i has value 0 in line ℓ . Then

1. $\hat{p}_1, \dots, \hat{p}_m \vdash F$ is valid if F evaluates to 1 in line ℓ
2. $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F$ is valid if F evaluates to 0 in line ℓ

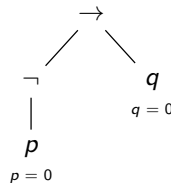
Proof by **Structural Induction** on structure of F , i.e., induction on height of parse tree for F .

- ▶ Base Case : If $F = p$, a proposition, then $p \vdash p$ and $\neg p \vdash \neg p$.
- ▶ Induction Hyp: Assume Lemma for formulae with parse tree height $\leq k - 1$
- ▶ Case Negation : $F = \neg F_1$
 - ▶ If F evaluates 1 at line ℓ , then F_1 evaluates to 0. By **inductive hypothesis**, $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F_1$, i.e., $\hat{p}_1, \dots, \hat{p}_m \vdash F$.
 - ▶ If F evaluates 0 at line ℓ , then F_1 evaluates to 1. By **inductive hypothesis**, $\hat{p}_1, \dots, \hat{p}_m \vdash F_1$. Use $\neg \neg i$ rule to obtain proof of $\hat{p}_1, \dots, \hat{p}_m \vdash \neg \neg F_1$, i.e., $\hat{p}_1, \dots, \hat{p}_m \vdash \neg F$.

Seeing the Induction: An Example

Let $F = \neg p \rightarrow q$, i.e., $F_1 = \neg p$ and $F_2 = q$. We derive the proof for line ℓ below.

p	q	$\neg p$	F	
0	0	1	0	$\leftarrow \ell$
0	1	1	1	
1	0	0	1	
1	1	0	1	



Goal: $\hat{p}, \hat{q} \vdash \neg F$.

- **Leaves:** $p = 0 \Rightarrow \hat{p} = \neg p$, so $\neg p \vdash \neg p$ (base case).
 $q = 0 \Rightarrow \hat{q} = \neg q$, so $\neg q \vdash \neg q$ (base case).
- **$\neg$ node ($F_1 = \neg p$):** F_1 evaluates to 1 $\Rightarrow$ using the leaf, $\neg p \vdash \neg p$, i.e., $\neg p \vdash F_1$.
- **$\rightarrow$ node (F , root):** F evaluates to 0 at this line. We have $\neg p$ (from the $\neg$ node) and $\neg q$ (from the leaf) as premises — using these, we derive $\neg F$ on the right:

- $\neg p$ premise
- $\neg q$ premise
- $\neg p \rightarrow q$ assumption
- q MP 3,1
- $\perp$ $\neg e$ 4,2
- $\neg(\neg p \rightarrow q)$ $\neg i$ 3-5